2025/1/8 Preliminary Examination - 2025

X Mathematics (Part - 1) (71)

Time 2 hours (Pages-5) Marks - 40

Instructions -

1) All questions are compulsory
2) Use of calculator is not allowed
3) Figures to the right side of questions indicate full marks.
4) Assessment will be done for the first attempted answer to MCQ (Q.1.A)
5) Write the correct alternative letter (A,B,C,D) only for MCQ.

Q.1A) Choose the correct answer and write the alphabet of it in front of the subquestion number (4)

1) To draw the graph of $2x - 6y = 3$ Find x when y = 1

A) 6 B) 9 C) 1 D) 4.5

2) Which of the following is a quadratic equation?

A) $\frac{1}{x} + 3 = x^2$ B) $x(x+5) = 2$ C) $n - 1 = 2n$ D) $\frac{1}{x^2}(x+2) = x$

3) For an A.P. a = 3, d = 5 then t_5 =

A) 23 B) 13 C) 18 D) 28

4) If n(A) = 3, $P(A) = \frac{3}{4}$ then n(S) = ?

A) 4 B) $\frac{3}{2}$ C) $\frac{4}{3}$ D) [illegible]

B) Solve the following subquestions (4)

1) Find the value of the following determinant

$$\begin{vmatrix} 5 & 3 \\ 2 & 4 \end{vmatrix}$$

2) If two coins are tossed simultaneously. Write the sample space 'S'

3) If rate of CGST on an article is 9%. What will be the rate of SGST?

4) Find common difference for an A.P. [illegible]

Q.2A) Complete any two activities of the following. (4)

1) To find the value of x complete the activity,

$3x - 4y = -7$ ---- equation (I) ; $5x - 2y = 7$ ---- equation (II)

On multiplying both sides of equation (II) by -2 , we get

$\boxed{} + 4y = -14$ ----equation (III)

Add equations (I) and (III)

$$\begin{array}{r} 3x - 4y = -7 \\ + \underline{\;-10x + 4y = -14} \\ \boxed{} = -21 \end{array}$$

$$\therefore\ x = \frac{-21}{\boxed{}} = \boxed{}$$

2) Complete the following activity to determine the nature of the roots of the quadratic equation $2x^2 - 7x + 2 = 0$

Activity : Complete $2x^2 - 7x + 2 = 0$ with $ax^2 + bx + c = 0$

$a = 2,\ b = \boxed{},\ c = 2$

$b^2 - 4ac = (-7)^2 - 4 \times 2 \times 2$

$= \boxed{} - 16$

$= 25$

$\therefore\ b^2 - 4ac > 25$

∴ The roots of the equation are real and $\boxed{}$

3) If 100 shares of FV ₹ 10 were purchased for MV of ₹ 250. Company declared 20% dividend on the shares then to find (i) sum investment (ii) Dividend received (iii) Rate of return. Complete the following activity.

(i) Sum investment $= 250 \times 100 =$ ₹ $\boxed{}$

(ii) Dividend per share $= \boxed{} \times \frac{20}{100} =$ ₹ 20

∴ Total dividend received on 100 shares = 100 x 20 = ₹2000

(iii) Rate of return $= \dfrac{\boxed{}}{\textit{Sum invested}} \times 100$

$= \dfrac{2000}{25{,}000} \times 100\ \boxed{}\%$

B) Solve the following subquestions (any four) **(8)**

1) The number of social media users in a building is given in the following table. Draw its pie-diagram.

Social Media	X	Facebook	Instagram	Whatsapp
Number of Users	40	35	05	20

2) Obtain the quadratic equation if roots are -3, -7.

3) Solve the simultaneous equations : $4x - 2y + 4 = 0$; $4x + 3y - 16 = 0$

4) Yami Beauty Products paid 18% GST on cosmetics worth ₹60,000 and sold to a customer for ₹ 100,000. What are the amounts of CGST and SGST shown in the tax invoice issued?

5) First term and common difference of an A.P. are 2 and 2 respectively; find S_{20}.

Q.3A) Complete any one activity and rewrite it. **(3)**

1) A road safety committee of 2 members is to be formed from two boys (B_1, B_2) and two girls (G_1, G_2) . To write sample space's', complete the following activity.

(i) A committee of two boys = $\boxed{\quad}$

(ii) A committee of two Girls = $\boxed{\quad}$

(iii) A committee must be of one boy and one girl = $\{B_1G_1, B_2G_1, B_1G_2, \boxed{\quad}\}$

(iv) ∴ Sample space 's' = $\{B_1B_2,\ B_1G_1, \boxed{\quad}, \boxed{\quad}, B_2G_2, \boxed{\quad}\}$

2) A manufacture sold a cycle for a taxable value of ₹ 4000 to the wholesaler. Wholesaler sold it to the retailer for ₹ 4800 (taxable value). Retailer sold to a customer for ₹ 5200 (taxable value). Rate of GST is 12%. Complete the following activity to find the payable CGST and SGST at each stage of trading.

Solution : Out put tax of manufacture = 12% of 4000 = $\boxed{\dots \times \frac{\dots}{\dots}} = \boxed{\quad}$

GST payable by manufacture = ₹ 480.

Out put tax of wholesaler = 12% of 4800 = ₹ 576

∴ GST payable by wholesaler = Out put tax - Input tax

= ₹ (576 - 480) = ₹ 96

Out put tax of retailer = 12% of 5200 = ₹ $\boxed{}$

∴ GST payable by retailer = Out put tax of retailer - ITC of retailer

= ₹ ($\boxed{}$ - $\boxed{}$) = ₹ $\boxed{}$

B) Solve the following sub - questions (any two) **(6)**

1) Solve the following simultaneous equations using Cramer's rule :

$5x - 6y = -30$, $5x + 4y = 20$

2) Solve the following quadratic equation by formula method :

$5y^2 + 5y = 1$

3) A box contains 30 tickets, bearing only one number from 1 to 30 on each. If one ticket is drown at random, find the probability of an event that the ticket drawn (i) an even number (ii) a complete square number.

4) In the given frequency distribution table of banana yield of banana growers, find the median of yield.

Generated (₹ thousand)	75-80	80-85	85-90	90-95	95-100
Number of banana growers	20	25	15	10	10

Q.4) Solve the following sub - questions (any two) **(8)**

1) If the 9th term of an A.P. is zero then show that the 29th term is twice the 19th term.

2) Sum of the present age of Dinesh and Pradnya is 106. and product of their ages is 2709. Find their present ages.

3) The percentage of std 10th result of a school for the year 2022-23 and the number of students is given in the table below. Draw a frequency polygon

Result %	30-40	40-50	50-60	60-70	70-80	80-90	90-100
Number of students	07	33	45	65	47	18	05

Q.5) Solve the following sub - questions (any one) (3)

1) Draw the graph of the equations $x + 4y = 7$; $2x - y = 5$. Identify the type of the triangle formed by the lines after intersecting each other (According to sides and angles) Write both the types.

2) Find the mean from the following table by 'Assumed mean' method

Daily wages (₹)	400-480	480-560	560-640	640-720	720-800
No. of workers	05	10	15	12	08